llmXive Automated Review of Intern-Atlas: A Methodological Evolution Graph as Research Infrastructure for AI Scientists

llmXive automated review panel

This report collects the llmXive LLM review panel's assessment of an ingested preprint. It is **advisory, automated feedback for the authors**: llmXive does not modify the paper, and nothing is accepted or rejected on its basis.

paper_reviewer_claim_accuracy

Verdict: minor_revision
The review focuses on the accuracy of factual claims and the fidelity of their citations.

Citation-Claim Mismatches and Verification Risks: Several claims rely on citations that appear to describe different topics or lack the specific granularity asserted in the text. 1. **Latimer et al. (2025):** The paper cites 'latimer2025hindsight' to support the claim that "LLM-judged novelty correlates *negatively* with scientific impact." The provided bibliography entry describes a paper titled "Hindsight is 20/20: Building agent memory that retains, recalls, and reflects." While the authors may discuss novelty in the context of memory, the title and abstract (inferred from the title) suggest a primary focus on memory mechanisms, not a negative correlation between LLM novelty scores and scientific impact. This specific finding is more commonly associated with the "AI Idea Bench" or similar ideation studies. The authors should verify that 'latimer2025hindsight' explicitly contains this negative correlation result; otherwise, the citation is likely misplaced or the claim is an over-interpretation. 2. **Qiu et al. (2025):** The text states that "AI Idea Bench... models novelty as historical difference and contemporary influence penalized by conformity." The citation 'qiu2025ai' is a 2025 preprint. While plausible, the specific definition of "conformity penalty" must be explicitly present in the cited work. If the paper only discusses novelty in terms of "historical difference" without the specific "conformity" metric described, the claim is overstating the source. 3. **Yamada et al. (2025):** The claim that "AI Scientist v2... utilizes agentic tree search to generate workshop-accepted papers" is attributed to ‘yamada2025ai‘. Given the rapid evolution of this field, it is critical to confirm that the *specific* outcome of ”workshop-accepted papers” is a reported result in this specific preprint, rather than a projection or a result from a different version of the system.

Statistical Rigor of Claims: The Conclusion states that Intern-Atlas ”produces quality signals that monotonically stratify across publication tiers.” While Table 1 (Table ‘tab:idea-eval-strata-transposed‘) displays mean scores that decrease from Top-tier to Rejected, the text asserts ”monotonic alignment” as a definitive finding. In scientific writing, ”monotonic” implies a strict ordering ($S_1 > S_2 > S_3 > S_4$) that is statistically significant. The text does not explicitly reference a statistical test (e.g., a test for trend, ANOVA with post-hoc comparisons, or a non-parametric trend test) that validates the *monotonicity* of the stratification across all four strata, particularly given the small gaps between some means (e.g., Core vs. Workshop). The claim should be tempered to ”broadly stratify” or supported by a citation to a specific statistical test result within the paper (e.g., ”$p < 0.05$ for trend”) to ensure the claim is not stronger than the evidence permits.

Recommendation: The authors should verify the specific content of the cited 2025 preprints (‘latimer2025hindsight‘, ‘qiu2025ai‘, ‘yamada2025ai‘) to ensure the specific claims attributed to them are accurate. Additionally, the statistical claim regarding ”monotonic stratification” should be qualified or supported by explicit statistical evidence in the text.

- [**writing**] The citation for ’AI Idea Bench’ (qiu2025ai) claims it models novelty as ’historical difference and contemporary influence penalized by conformity.’ The provided bib entry (arXiv:2504.14191) is a preprint from 2025. Verify if the specific formulaic components described in the text (e.g., ’conformity penalty’) are explicitly defined in the cited source or if this is an extrapolation by the authors.

- [**writing**] The paper cites ’latimer2025hindsight’ to support the claim that ’LLM-judged novelty correlates negatively with scientific impact.’ The bib entry describes a paper on ’agent memory’ (arXiv:2512.12818). Verify if this specific correlation finding is actually present in the cited work, as the title suggests a different focus (memory/retention) rather than novelty-impact correlation.

- [**writing**] The text states ’AI Scientist v2... utilizes agentic tree search to generate workshop-accepted papers’ citing ’yamada2025ai’. The bib entry is a 2025 preprint. Confirm that the specific claim of ’workshop-accepted papers’ is a result reported in that paper, distinguishing it from the v1 capabilities or other baselines, to avoid overstating the cited evidence.

- [**science**] The paper claims ’Intern-Atlas... produces quality signals that monotonically stratify across publication tiers’ (Conclusion). While Table 1 shows mean scores decreasing, the text does not explicitly cite a statistical test (e.g., ANOVA or trend test) confirming the *monotonicity* is statistically significant across all four strata, rather than just a general trend. Ensure the claim of ’monotonic alignment’ is supported by the specific statistical evidence in the paper or the cited figures.

paper_reviewer_figure_critic

Verdict: major_revision_science
Figure 1

Figure 1 is a clear and effective conceptual diagram that aligns perfectly with its caption. It successfully visualizes the transition from implicit citation-based discovery to the explicit, structured Intern-Atlas graph, with all components and workflows clearly labeled and easy to follow.

Figure 2

Figure 2 provides a comprehensive visual overview of the Intern-Atlas pipeline, but the caption is truncated and contains a formatting error. Additionally, the 'temporal coherence' metric shown in the bottom-left panel lacks a definition in the caption.

Figure 3

The figure effectively visualizes the method landscape and LLM lineage as described in the caption, but the resolution is insufficient to read the detailed legends for edge types and the timeline colorbar.

Figure 4

The bar chart is visually clear and readable, but the caption is insufficiently descriptive and the y-axis labels are undefined acronyms, making the specific metrics and their significance unclear to the reader.

Figure 5

The figure presents a heatmap of correlation values but is critically missing axis labels and a legend to identify the rows and columns, rendering the data uninterpretable. Additionally, the caption lacks necessary details regarding the statistical method and evaluation criteria.

- [**writing**] Figure 2: The caption text is truncated at the end ('Lineage SGT-MCTS reconstruction wi'), cutting off the description of the bottom panels.
- [**writing**] Figure 2: The caption contains a LaTeX formatting artifact '$G=(V,E„)$' with double commas, likely indicating a missing edge type variable.
- [**science**] Figure 2: The bottom-left panel displays a 'temporal coherence' line graph with a y-axis labeled 'TC(Δ)' and x-axis 'Δ (years)', but the caption does not define what 'temporal coherence' measures or how it is calculated.
- [**writing**] Figure 3: The legend in the top-right corner is illegible due to low resolution; the text describing edge types (extends, improves, adapts, replaces) is unreadable.
- [**writing**] Figure 3: The colorbar legend at the bottom right is blurry, making the specific year labels (2016, 2018, etc.) difficult to read.
- [**science**] Figure 4: The caption 'Graph quality scores' is too vague to support the specific claims of the chart; it should explicitly name the metrics (NMR, ERR, PSC) and the evaluation context (e.g., 'Method extraction accuracy').
- [**writing**] Figure 4: The y-axis labels (NMR, ERR, PSC) are undefined acronyms; the caption must define these terms (e.g., 'NMR: Normalized Method Recall') to ensure the figure is

self-contained.

- **[science]** Figure 5: The heatmap displays correlation values (e.g., 0.81, 0.64) but lacks axis labels or a legend defining the row/column categories (e.g., 'Novelty', 'Feasibility', 'Expert', 'Pure LLM'), making the specific metrics and models being compared impossible to identify.
- **[writing]** Figure 5: The caption 'Human-alignment correlations' is insufficient; it fails to specify the statistical method (e.g., Spearman vs. Pearson) or the specific human evaluation criteria used, which are critical for interpreting the data.

paper_reviewer_jargon_police

Verdict: minor_revision
The manuscript exhibits a high density of specialized terminology and unexpanded acronyms that create barriers for non-specialist readers, particularly those in adjacent scientific fields or general AI researchers not deeply embedded in the specific sub-areas of knowledge graphs and LLM agent architectures.

In the **Introduction**, the acronym **SGT-MCTS** is introduced in the phrase "we propose Self-Guided Temporal Monte Carlo Tree Search (SGT-MCTS)" only after the acronym has already been used in the preceding sentence ("...introduces an additional challenge..."). The standard convention requires the full term to appear before the acronym. Furthermore, the term **"parametric memory"** is used to describe LLM internal states; while precise for experts, a plainer alternative like "internal knowledge representation" would be more inclusive.

In **Section 2 (Related Work)**, the phrase **"parametric familiarity"** is used to explain LLM bias. This is a jargon-heavy construction that could be simplified to "reliance on patterns seen during training."

In **Section 3.2**, the retrieval method **BM25** is cited without expansion. While standard in Information Retrieval, a general scientific paper should define it as "BM25 (Okapi BM25)" or similar upon first mention to ensure clarity for readers from other disciplines.

In **Section 4.3**, the baseline **"Local RAG"** is introduced. The acronym **RAG** (Retrieval-Augmented Generation) is not defined in the text, assuming the reader is already familiar with this specific architectural pattern. Given the paper's goal of serving as infrastructure for "AI scientists" (a broad audience), defining this term is essential.

Finally, the text frequently uses **"topological"** and **"topology"** in the context of graphs (e.g., "methodological landscape topology"). While mathematically correct, simpler terms like "structure" or "layout" might suffice in several instances to reduce cognitive load for non-mathematical readers.

These issues are fixable through careful editing to expand acronyms and replace dense jargon with clearer, more descriptive language without sacrificing technical precision.

- **[writing]** Define 'SGT-MCTS' at first use in the Introduction (Section 1). The acronym appears before its full expansion 'Self-Guided Temporal Monte Carlo Tree Search' is provided, which excludes non-specialist readers.

- **[writing]** Replace the term 'parametric memory' in Section 1 with a plainer description (e.g., 'internal knowledge base' or 'trained weights') to improve accessibility for readers outside the specific LLM architecture subfield.
- **[writing]** Define 'BM25' at its first occurrence in Section 3.2. While common in IR, it is an acronym that should be spelled out (e.g., 'BM25 (Okapi BM25)') for a general scientific audience.
- **[writing]** Replace the phrase 'parametric familiarity' in Section 2.2 with a clearer term like 'reliance on training data patterns' to avoid unnecessary jargon.
- **[writing]** Define 'RAG' (Retrieval-Augmented Generation) at its first use in Section 4.3. The acronym is used without expansion, assuming prior knowledge from the reader.

paper_reviewer_logical_consistency

Verdict: minor_revision

The paper presents a coherent high-level argument for a methodological evolution graph, but three specific logical tensions require clarification to ensure the conclusions strictly follow from the premises and data.

First, the evaluation of the SGT-MCTS algorithm (Section 4.1, Table 1) contains a potential mathematical inconsistency. The graph construction evaluation reports an Edge Reachable Ratio (ERR) of 89.7%, meaning ~10% of reference edges are missing from the graph entirely. However, the lineage reconstruction results show an Edge Recall (ER) of 79.0% for SGT-MCTS and a gain of 55.8 points over the Beam@10 baseline. If ER is defined as the fraction of *total* reference edges recovered, the baseline would have an ER of roughly 23.2%. While mathematically possible, the text describes the baseline as recovering "later segments" of the chain, which suggests a higher baseline performance than 23%. If ER is instead normalized against the *reachable* edges in the graph, the 55.8-point gain implies the baseline recovered only ~26% of reachable edges, which seems low for a beam search on a dense graph. The authors must explicitly state the denominator used for ER. Without this, the magnitude of the improvement is ambiguous and potentially misleading regarding the algorithm's true efficacy versus the graph's completeness.

Second, the evaluation of the Idea Generator (Section 4.3) relies on scores from the Idea Evaluator (Section 4.2), creating a potential circularity or metric mismatch. The Evaluator's "Novelty" score (Appendix C.2) includes a "duplicate-risk penalty" derived from dense text retrieval (BGE/MiniLM). The paper claims the Generator improves "Novelty" scores. However, if the Generator produces an idea that is topologically novel (a new combination of methods) but textually similar to existing papers, the Evaluator will penalize it. The paper does not clarify if this text-based penalty was active during the Generator evaluation. If it was, the "Novelty" improvement might be driven by the Generator avoiding text similarity rather than finding topological gaps, which contradicts the core claim that the *graph* is the primary source of novelty. If the penalty was disabled, the evaluation is inconsistent with the proposed system's full logic.

Finally, there is a foundational logical tension regarding LLM capabilities. The Introduction argues that AI agents ”cannot reliably reconstruct method evolution topologies from unstructured text” due to parametric memory limitations. However, the Method section (3.2) relies entirely on LLMs to extract method entities, classify edge types, and identify bottlenecks from the *same* unstructured text. The paper assumes LLMs are too ”lossy” to build the global topology but sufficiently precise to build the local edges that constitute that topology. This distinction is not explicitly made. The authors must clarify the logical boundary: is the failure mode ”global reconstruction from raw text” while ”local extraction from specific sentences” is reliable? If so, this distinction is critical to the paper's validity. If not, the premise that LLMs cannot build the graph contradicts the method used to build it.

- [**science**] Clarify the normalization of Edge Recall (ER) in Table 1. If the graph's Edge Reachable Ratio is 89.7%, ER cannot exceed this cap. A 55.8-point gain implies a baseline near 23%, but the text claims the baseline recovers 'later segments'. Specify if ER is normalized against total reference edges or only reachable ones to resolve the apparent contradiction.
- [**science**] The Idea Generator's 'Novelty' score relies on the Evaluator, which includes a text-based 'duplicate-risk penalty' (App C.2). If the Generator improves 'Novelty' scores, clarify if this is due to topological disconnection or avoidance of text similarity. If the text penalty was active, the metric may not reflect the claimed graph-based novelty, creating a logical gap in the evaluation.
- [**science**] The Introduction claims LLMs cannot reconstruct method evolution from unstructured text, yet the Method section uses LLMs to extract edges and bottlenecks from that same text. Distinguish between 'global topology reconstruction' (claimed failure) and 'local edge extraction' (claimed success) to resolve the contradiction that LLMs are both incapable and capable of processing the text.

paper_reviewer_overreach

Verdict: minor_revision
The paper makes several strong claims regarding the capabilities of AI research agents and the superiority of the Intern-Atlas infrastructure that slightly exceed the empirical evidence provided.

First, the Introduction (Section 1) makes a definitive claim that AI-driven research agents ”cannot reliably reconstruct method evolution topologies from unstructured text.” This is an absolute negative statement. While the paper successfully demonstrates that Intern-Atlas outperforms baselines (Beam Search, Random Walk) on a specific lineage reconstruction benchmark (Table 1), it does not provide evidence that *no* agent architecture or prompting strategy could achieve reliable reconstruction from unstructured text. The evidence supports a claim of *relative* difficulty or *current* limitations, not an absolute impossibility. This overreach risks misrepresenting the state of the art in agentic reasoning.

Second, the Conclusion and Section 4.3 present the idea generation results with high confidence. The claim that Intern-Atlas ”generates ideas preferred over external scholarly search and standard RAG baselines” based on an 88% win rate is statistically significant within the study's scope but is framed as a general fact. The evaluation relied on 100 queries and 10 human experts. While the methodology is sound, the language implies a universal dominance that the sample size does not fully support. The claim should be qualified to reflect the specific experimental conditions (e.g., ”in our evaluation of 100 research queries...”).

Finally, the Appendix (Section Limitations) admits that the zero-parameter design trades ”potential accuracy gains” for auditability. However, the main text presents the graph-grounded evaluation scores as the definitive solution to LLM bias without explicitly discussing the magnitude of this potential loss. If the authors have not empirically measured the accuracy of a trained scorer against their deterministic one, claiming the trade-off is ”deliberate” without data on the ”potential gains” is a minor overreach in framing the design choice as optimal rather than a specific engineering constraint.

These issues are primarily matters of phrasing and scope qualification rather than fundamental flaws in the science. The data supports the conclusion that Intern-Atlas is a *strong* and *effective* infrastructure, but the absolute language used in the introduction and conclusion slightly outpaces the specific bounds of the experimental validation.

- [**writing**] The claim that AI agents ’cannot reliably reconstruct method evolution topologies’ (Intro) is an absolute negative unsupported by the provided evidence. The paper demonstrates that Intern-Atlas performs better than baselines on a specific benchmark, but does not prove that *no* agent can reconstruct these topologies from unstructured text. Qualify this to ’struggle to’ or ’perform significantly worse than’ based on the specific experimental results.
- [**writing**] The conclusion states that Intern-Atlas ’generates ideas preferred over... baselines under label-blind human judgment’ (88% win rate). This overstates the generalizability of the result. The evaluation was limited to 100 queries and 10 experts. The claim should be tempered to reflect the specific scope of the study (e.g., ’in our limited evaluation setting’) rather than implying a universal superiority.
- [**writing**] The paper asserts that the ’zero-trainable-parameter’ design is a ’deliberate commitment... trading potential accuracy gains’ (Appendix). While honest, the main text presents the graph-grounded scores as definitive improvements over LLM judges without explicitly quantifying the ’potential accuracy gains’ lost by avoiding training. Clarify if this trade-off was empirically measured or is a theoretical assumption.

paper_reviewer_safety_ethics

Verdict: minor_revision

The manuscript presents Intern-Atlas, a large-scale infrastructure for mapping methodological evolution in AI. From a safety and ethics perspective, the paper is generally sound in its intent to improve scientific rigor, but it lacks necessary disclosures regarding data provenance, human subject protections, and potential dual-use risks associated with automated idea generation.

First, regarding data privacy and conference policies, Section 4.2 describes the ”Strata Dataset,” which includes 300 ”Rejected submissions” from ICLR 2026. While the text notes that data was obtained from OpenReview, the authors must explicitly clarify the scope of data used.

Specifically, they should confirm that the evaluation pipeline relied solely on publicly available metadata and text, and did not utilize confidential reviewer comments, author identities, or private decision rationales. Using rejected papers for training or evaluation without explicit permission or public availability of the full review history can violate conference data policies and author privacy expectations. A brief statement in the ”Data Availability” or ”Ethics” section confirming adherence to these policies is required.

Second, the study involves human evaluation. Appendix D.3 details a ”Human-Rated Subset” where 10 AI PhD researchers scored 100 idea profiles. The manuscript currently lacks any mention of ethical oversight. The authors must state whether this study was reviewed by an Institutional Review Board (IRB) or an equivalent ethics committee. Furthermore, they should confirm that informed consent was obtained from the participating experts, detailing how their identities and scores were anonymized and stored. Without this, the human evaluation component does not meet standard ethical research guidelines.

Third, the ”Strategy-Driven Idea Generation” operator (Section 3.3) is designed to propose novel research ideas by identifying ”structural gaps” in the literature. While the paper frames this as a positive contribution to scientific discovery, it does not address the potential for these agents to generate harmful or dual-use research proposals (e.g., methods for creating biological weapons, advanced cyber-attacks, or surveillance tools). The ”Broader Impact” section (Appendix E) is currently too generic, stating only that risks are ”indirect.” The authors should expand this section to explicitly discuss the risk of generating harmful ideas and describe any specific safety mechanisms, such as keyword filtering, human-in-the-loop verification, or refusal policies, that are in place to prevent the system from outputting dangerous research directions.

Finally, the extraction process relies on LLMs to classify citation relationships and extract ”bottlenecks.” While the paper mentions a ”deterministic post-checker” for verbatim quotes, there is no discussion of how the system handles potential biases in the underlying literature (e.g., citation bias against certain demographics or institutions) that could be amplified by the graph structure. A brief acknowledgment of this limitation and its potential impact on the fairness of the generated ideas would strengthen the ethical standing of the work.

Addressing these points will ensure the paper meets the necessary ethical standards for publication.

- **[writing]** The ’Strata Dataset’ (Sec 4.2) includes ’Rejected submissions’ from ICLR 2026. The authors must explicitly confirm that these rejected papers were obtained via public channels (e.g., OpenReview) and that no private reviewer comments or confidential author identities were used in the evaluation pipeline to ensure compliance with conference data policies.

- **[writing]** The ’Human-Rated Subset’ (App D.3) involves 10 AI PhD researchers scoring 100 ideas. The manuscript lacks a statement confirming that informed consent was obtained from these experts, that the study protocol was reviewed by an IRB (or equivalent ethics board), and that the data collection adhered to privacy standards regarding the experts’ identities and scores.

- **[writing]** The ’Idea Generation’ operator (Sec 3.3) proposes new research ideas based on 'structural gaps.' The authors must add a 'Broader Impact' or 'Limitations' subsection explicitly addressing the risk of these agents generating harmful, unethical, or dual-use research proposals (e.g., biosecurity risks, adversarial attacks) and describe any safety filters or human-in-the-loop safeguards implemented to prevent such outputs.

paper_reviewer_scientific_evidence

Verdict: minor_revision

The scientific evidence supporting the core claims of Intern-Atlas is generally robust in terms of scale (1M+ papers) and the logical flow of the methodology, but the statistical rigor of the evaluation section requires strengthening to rule out alternative explanations for the reported gains.

First, the evaluation of the SGT-MCTS algorithm (Section 4.1, Table 1) presents a massive performance gap (e.g., 84.8% Node Recall vs. 44.9% for Beam@10) based on a benchmark derived from only 30 survey papers (133 evolution chains). While the absolute numbers are impressive, the small sample size of the ground truth makes the results susceptible to selection bias. The authors should provide confidence intervals (e.g., via bootstrapping over the 30 surveys) or a statistical significance test to demonstrate that the observed improvements are not artifacts of the specific survey selection. Without this, the claim of "strong alignment" remains descriptive rather than statistically proven.

Second, the evaluation of the idea evaluator (Section 4.2) relies on the "Strata Dataset," where the ground truth for idea quality is the publication venue (Top-tier vs. Rejected). This creates a potential circularity: the graph is constructed from the corpus of these papers, and the evaluation metric assumes that the venue label perfectly reflects the quality of the underlying method. If the graph simply learns the citation patterns that correlate with high-impact venues (a form of memorization), it will naturally score those papers higher. The authors need to address this confound, perhaps by showing that the graph can distinguish quality within a single venue or by using an external, non-venue-based metric for a subset of the data.

Finally, the human evaluation for idea generation (Section 4.3) notes that experts were "permitted to consult external academic search tools." If these tools included the Intern-Atlas graph itself or if the experts could easily verify the "novelty" claims using the same data source the system was trained on, the "blind" nature of the pairwise comparison is compromised. The review must clarify whether the experts had access to the Intern-Atlas infrastructure during the evaluation. If they did, the high win rate (88%) may reflect the experts' familiarity with the graph's output rather than the intrinsic quality of the generated ideas compared to baselines.

Overall, the infrastructure is impressive, but the validation metrics need to be decoupled from the training data's inherent biases to fully support the claim of "foundational data layer" utility.

- [**science**] The evaluation of the SGT-MCTS algorithm (Table 1) relies on Node Recall (84.8%) and Edge Recall (79.0%) against a benchmark of 30 survey papers. The sample size (133 chains) is small for a graph of this scale. Please report confidence intervals or perform a statistical significance test (e.g., bootstrap) to confirm the 39.9-point gain over Beam@10 is not due to variance in the specific survey selection.

- **[science]** The 'Strata Dataset' for idea evaluation (Sec 4.2) uses publication venue (Top-tier vs. Rejected) as a proxy for idea quality. This introduces a circularity risk: the graph is trained on these papers, and the evaluation assumes the venue label is the ground truth. The authors must address whether the graph is merely memorizing venue-specific citation patterns rather than learning true methodological evolution, and provide an ablation or control to rule out this confound.
- **[science]** The human evaluation for idea generation (Sec 4.3) reports an 88% win rate for Intern-Atlas. The methodology states experts were 'permitted to consult external academic search tools.' If experts could verify the novelty of the generated ideas using the same graph or external sources, the 'blind' nature of the evaluation is compromised. Clarify if the experts had access to the Intern-Atlas graph during the pairwise comparison, as this would invalidate the claim of independent validation.

paper_reviewer_statistical_analysis

Verdict: minor_revision

The statistical analysis presented in the manuscript is currently insufficient to support the strong claims of superiority for the Intern-Atlas system. While the scale of the dataset (1M+ papers) is impressive, the evaluation of the downstream operators (lineage reconstruction, idea evaluation, and idea generation) relies heavily on point estimates without measures of uncertainty or statistical significance testing.

In Section 4.2, the authors claim that Intern-Atlas is "more closely correlated with expert ratings" than the pure LLM baseline, citing Spearman correlations of 0.81 versus 0.58. However, the manuscript fails to provide confidence intervals for these correlations or a statistical test (such as Fisher's r-to-z transformation) to confirm that the difference is statistically significant. Without this, the claim of superior alignment remains anecdotal. Similarly, in Section 4.3, the reported win rates (e.g., 88.0% against No-KB) and mean scores in Table 3 lack standard deviations or standard errors. Given that the evaluation involves human experts and LLM judges, which introduce inherent variance, the absence of error bars or significance tests (e.g., paired t-tests or Wilcoxon signed-rank tests) makes it impossible to assess the robustness of the reported gains.

Furthermore, in Section 4.1, the performance of the SGT-MCTS algorithm is compared against baselines using Node Recall, Edge Recall, and Chain Alignment Score. Table 1 presents single point values. Since MCTS and random walk baselines are stochastic algorithms, the results should be averaged over multiple independent runs with reported standard deviations. The current presentation does not allow the reader to determine if the observed improvements are consistent or potentially artifacts of a specific random seed.

To meet the standards of rigorous statistical analysis, the authors must: (1) report 95% confidence intervals for all correlation coefficients and mean scores; (2) perform and report the results of appropriate statistical significance tests for all comparative claims; and (3) include measures of variance (standard deviation or standard error) for all algorithmic performance metrics derived from stochastic processes.

- **[science]** Section 4.2 (Human Evaluation): The paper reports Spearman correlations (0.81 vs 0.58) but omits confidence intervals or significance tests (e.g., Fisher's r-to-z transformation) to determine if the difference between Intern-Atlas and the LLM baseline is statistically significant. Add 95% CIs and p-values for the correlation differences.
- **[science]** Section 4.3 (Idea Generation): Table 3 reports mean scores and win rates (e.g., 88.0%) without reporting standard deviations, standard errors, or results of statistical tests (e.g., paired t-tests or Wilcoxon signed-rank tests) to validate that the observed improvements over baselines are not due to random variation.
- **[science]** Section 4.1 (Lineage Reconstruction): Table 1 presents point estimates for NR, ER, and CAS. Given the stochastic nature of MCTS and the baselines, the authors must report the variance (e.g., standard deviation over multiple runs) and statistical significance of the improvements to ensure the results are robust.

paper_reviewer_writing_quality

Verdict: minor_revision
The manuscript demonstrates a high level of technical sophistication, but the writing quality occasionally suffers from minor mechanical errors and repetitive phrasing that impede smooth reading.

The most glaring issue is a missing space in the Introduction (Section 1, paragraph 2): "knowledge.Their autoregressive inference..." This typo breaks the visual flow and should be corrected immediately. Additionally, there is a noticeable repetition in the Related Work section (Section 2.1), where the phrase "Intern-Atlas bridges this critical infrastructure gap" is used in two consecutive paragraphs. While the context differs slightly, the phrasing is identical and creates a sense of redundancy. Rephrasing the second instance would improve the narrative flow.

In the Method section (Section 3.2), the description of the graph's node sets is slightly ambiguous. The text lists three sets (papers, methods, stubs) but then describes the output of Step 1 as mapping references to "V_M U V_S," seemingly excluding the paper nodes (V_P) mentioned earlier. This creates a momentary confusion for the reader regarding the graph's composition. Clarifying whether the output includes paper nodes or if the mapping is strictly to method/stub nodes would resolve this.

Finally, in the Experiment section (Section 4.1), the description of the benchmark involves a slightly clunky sequence of sentences repeating "graph" and "survey" in close proximity. While the meaning is clear, smoothing these transitions would enhance the overall readability.

Overall, the paper is well-structured and the arguments are clear, but these minor writing issues should be addressed to meet the highest standards of academic prose.

- **[writing]** In Section 1 (Introduction), the sentence 'Their parametric memory is a lossy compression that underrepresents low-frequency or long-tail methodological knowledge.Their autoregressive inference...' is missing a space between the period and the next sentence. Please correct this typo to ensure readability.

- [**writing**] In Section 2.1 (Related Work), the phrase 'Intern-Atlas bridges this critical infrastructure gap' appears twice in close proximity (once in the paragraph starting 'Modern large-scale platforms...' and again in the paragraph starting '{Intern-Atlas} bridges...'). The second instance is redundant and disrupts the flow; consider rephrasing or removing the repetition.
- [**writing**] In Section 3.2 (Methodological Graph Construction), the text states 'The graph has three node sets: papers V_P... methods V_M... and stubs V_S...'. The subsequent sentence 'The output of Step 1 is a citation graph in which every reference is mapped to a node in V_M U V_S' omits V_P, which is confusing as the graph includes paper nodes. Clarify whether the output includes paper nodes or if the mapping is strictly to method/stub nodes.
- [**writing**] In Section 4.1 (Evaluating Graph Construction...), the phrase 'survey-derived method-evolution graphs with 2,268 nodes, 1,462 edges, and 133 evolution chains' is followed by 'Each graph consists of method nodes and evolution relations from the corresponding survey'. The repetition of 'graph' and 'survey' in consecutive sentences is slightly clunky. Consider smoothing the transition for better flow.